Campus Circular Economy and Zero-Waste Policy Guidelines

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CAMPUS CIRCULAR ECONOMY AND ZERO-WASTE POLICY GUIDELINES

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Promulgated by: Campus Environmental Cell and Directorate of Facilities (DEP_ADMN).

Institutional Policy Target: TRG_ZERO_WASTE (Zero Waste to Landfill & 85% Recycling by 2028).

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. STATUTORY RATIONALE AND POLICY GOALS

Vivekananda Education Society's Institute of Technology (VESIT Green Campus) in Chembur, Mumbai, adopts these Circular Economy and Zero-Waste Guidelines to eliminate the generation of unsegregated solid refuse across its 32,500 m2 built estate. Serving a population of 3,200 students, faculty, and administrative staff, the college generated 68,400 kg of solid waste in 2024. Under baseline conditions, sending unsegregated municipal solid waste to Deonar landfill results in severe greenhouse gas emissions (EF_WASTE_LANDFILL = 1.25 kg CO2e/kg waste) and regional environmental contamination.

This policy establishes strict institutional mandates to divert 85.0% of solid waste from landfills by December 31, 2028 (TRG_ZERO_WASTE), supporting United Nations SDG 11 (Sustainable Cities and Communities) and SDG 12 (Responsible Consumption and Production).

2. MANDATORY FOUR-STREAM SOURCE SEGREGATION PROTOCOL

Effective September 1, 2024, all academic departments—Computer Engineering (DEP_CMPN), Information Technology (DEP_INFT), Electronics & Telecommunication (DEP_EXTC), AI & Data Science (DEP_AIDS), and Facilities (DEP_ADMN)—must maintain color-coded four-stream sorting receptacles at every corridor intersection, classroom block, and laboratory:

- Green Receptacle (Biodegradable Organics): Exclusively for wet food scraps, tea leaves, fruit peels, and floral waste from Canteen (BLD_CAN) and departmental lounges. Strictly zero plastic packaging. All green-bin contents are routed directly to the on-site anaerobic biomethanation plant and composting tumblers under project INIT_BIOGAS_CANTEEN.

- Blue Receptacle (Clean Dry Paper & Cardboard): Reserved for examination booklets, discarded textbooks, stationery paper, and corrugated cardboard boxes. Paper is bundled weekly and transferred to authorized municipal recyclers, delivering avoided carbon factor credits (EF_WASTE_RECYCLE = -0.65 kg CO2e/kg waste).

- Red Receptacle (Rigid Recyclable Plastics & Metal Cans): Designated for rigid polyethylene containers, clean aluminum beverage cans, and polypropylene labware.

- Black Receptacle (Inert Refuse & Sanitary Waste): Limited to sweepings, broken glass (wrapped separately), and non-recyclable sanitary refuse collected by municipal sanitation teams.

3. TOTAL PROHIBITION ON SINGLE-USE PLASTICS

In strict adherence to Maharashtra Plastic Waste Management rules:

- Prohibited Items: Single-use plastic cutlery, polystyrene food boxes, plastic straws, disposable plastic water bottles under 500ml, plastic promotional banners, and non-biodegradable parcel packaging are banned from campus grounds.

- Enforcement: Campus security conducts random baggage screening at entry gates. Commercial food vendors operating in the Canteen (BLD_CAN) face immediate financial penalties and contract revocation for supplying prohibited plastics.

- Alternative Mandates: Events must utilize reusable stainless steel cutlery, ceramic mugs, and certified compostable sugarcane bagasse containers.

4. SUSTAINABLE PROCUREMENT AND RECYCLED PAPER MANDATE

Institutional purchasing protocols must uphold circularity:

- Exam Paper Specifications: Administrative procurement must mandate that 100% of student semester examination answer booklets, lab journals, and administrative office stationery contain a minimum of 70% post-consumer recycled paper fiber.

- E-Waste Stewardship: Decommissioned computing hardware, monitors, and circuit boards from Computer Laboratories (BLD_LAB) must undergo formal material reclamation through certified e-waste partner Greenscape Solutions under E-Waste Rules 2022, ensuring complete recovery of copper, gold, and aluminum without landfill disposal.

Quarterly waste diversion audits published via the EcoClock AI dashboard shall verify departmental compliance against target TRG_ZERO_WASTE.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_WST_018**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*